

Exclusion and Visibility Geometry in VibeVolts

Simulation Reference Documentation

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1 Introduction

This document details the geometric, vector, and angular calculations used in VibeVolts to determine whether a target point in space is visible to a satellite or ground-based observatory detector.

Visibility is subject to two conditions:

1. **Field-of-View (FOV) Check:** The target must lie within the sensor's conical field-of-view.
2. **Exclusion Checks:** The detector's view must not be obstructed by, or lie too close to, key celestial bodies (Sun, Moon, Earth).

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2 Step 1: Coordinate Frame and Vectors

Let $\mathbf{x}_{\text{sat}}$ be the position of the observer (satellite or observatory) and $\mathbf{x}_{\text{targ}}$ be the position of the target, both represented in the Geocentric Celestial Reference System (GCRS).

The pointing vector from the observer to the target is:

$$\mathbf{v}_{\text{targ}} = \mathbf{x}_{\text{targ}} - \mathbf{x}_{\text{sat}} \quad [\text{meters}] \quad (1)$$

The unit pointing vector is:

$$\hat{\mathbf{v}}_{\text{targ}} = \frac{\mathbf{v}_{\text{targ}}}{\|\mathbf{v}_{\text{targ}}\|} \quad (2)$$

If the distance $\|\mathbf{v}_{\text{targ}}\| < 10^{-9}$ meters, the target is considered co-located with the satellite and is automatically marked as excluded.

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3 Step 2: Field-of-View (FOV) Cone Check

Let $\hat{\mathbf{p}}$ be the unit vector representing the pointing direction of the detector, and let FOV be the full-width angular diameter of the field-of-view in radians.

The angle θ_{targ} between the detector pointing vector and the target direction is:

$$\theta_{\text{targ}} = \arccos(\hat{\mathbf{p}} \cdot \hat{\mathbf{v}}_{\text{targ}}) \quad [\text{radians}] \quad (3)$$

The target is within the detector's field of view if and only if:

$$\theta_{\text{targ}} < \frac{\text{FOV}}{2} \quad (4)$$

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4 Step 3: Apparent Angular Radii of Celestial Bodies

As viewed from the observer's position $\mathbf{x}_{\text{sat}}$, celestial bodies appear as disks with angular sizes that depend on their physical radii and their distance from the observer.

Let $\mathbf{x}_{\text{body}}$ be the GCRS position of a celestial body (Sun, Moon, or Earth) and R_{body} be its physical radius. The vector from the observer to the body is:

$$\mathbf{v}_{\text{body}} = \mathbf{x}_{\text{body}} - \mathbf{x}_{\text{sat}} \quad (5)$$

And the unit vector is $\hat{\mathbf{v}}_{\text{body}} = \mathbf{v}_{\text{body}} / \|\mathbf{v}_{\text{body}}\|$.

The apparent angular radius θ_{body} of the body as seen by the observer is:

$$\theta_{\text{body}} = \arctan \left(\frac{R_{\text{body}}}{\|\mathbf{v}_{\text{body}}\|} \right) \quad [\text{radians}] \quad (6)$$

In VibeVolts, the physical parameters and apparent radii calculations are:

- **Sun:** Modeled as a point source for simplicity:

$$\theta_{\text{Sun}} = 0.0 \quad [\text{radians}] \quad (7)$$

- **Moon:** Using physical Moon radius $R_{\text{Moon}} \approx 1,737,400$ m:

$$\theta_{\text{Moon}} = \arctan \left(\frac{R_{\text{Moon}}}{\|\mathbf{v}_{\text{Moon}}\|} \right) \quad (8)$$

- **Earth:** Center is at GCRS origin $[0, 0, 0]$, so $\mathbf{v}_{\text{Earth}} = -\mathbf{x}_{\text{sat}}$. Using equatorial radius $R_{\text{Earth}} \approx 6,378,137$ m:

$$\theta_{\text{Earth}} = \arctan \left(\frac{R_{\text{Earth}}}{\|\mathbf{x}_{\text{sat}}\|} \right) \quad (9)$$

5 Step 4: Exclusion Angle Check

To protect sensitive optics from bright sources, a target observation is excluded if the line of sight passes too close to the limb (edge) of a celestial body.

Let ψ_{body} be the angular separation between the target direction and the celestial body center:

$$\psi_{\text{body}} = \arccos(\hat{\mathbf{v}}_{\text{targ}} \cdot \hat{\mathbf{v}}_{\text{body}}) \quad [\text{radians}] \quad (10)$$

Let θ_{limit} be the exclusion safety angle specified for the detector (e.g. `solarEx`, `lunarEx`, `earthEx` in radians). The angle between the target line-of-sight and the limb of the body is $\psi_{\text{body}} - \theta_{\text{body}}$.

The view is ****excluded**** (obstructed or blinded) if:

$$\psi_{\text{body}} - \theta_{\text{body}} < \theta_{\text{limit}} \quad (11)$$

If this inequality holds for the Sun, Moon, or Earth, the target is excluded from observation.

6 Summary of Variable Definitions

Variable	Code Representation	Description
$\mathbf{x}_{\text{sat}}$	<code>sat_pos / satposition</code>	Observer GCRS position vector (m)
$\mathbf{x}_{\text{targ}}$	<code>targets</code>	Target GCRS position vector (m)
$\hat{\mathbf{p}}$	<code>pointing</code>	Sensor pointing direction unit vector
FOV	<code>fov</code>	Sensor full field-of-view (rad)
$\mathbf{v}_{\text{body}}$	<code>vecs_to_bodies</code>	Vector from observer to celestial body center (m)
θ_{body}	<code>apparent_radii</code>	Apparent angular radius of celestial body (rad)
θ_{limit}	<code>solarEx, lunarEx, earthEx</code>	Detector exclusion safety angles (rad)
ψ_{body}	<code>angles</code>	Separation angle between target and body center (rad)

Table 1: Exclusion parameter and code variable mappings.